Customer Churn Analysis Project Report

1. Project Overview

This project focuses on analyzing customer churn in a telecom company. Customer churn refers to the situation where customers stop using a company's service. The goal of this project is to analyze customer data, identify patterns related to churn, and generate insights that can help businesses reduce customer loss. The analysis was performed using SQL for querying the dataset and Python for data cleaning and exploratory data analysis.

2. Dataset

The dataset used in this project is the Telco Customer Churn dataset. It contains information about telecom customers such as demographic details, services subscribed, billing information, and whether the customer has churned or not.

Key columns in the dataset include:

- customerID – Unique ID for each customer
- gender – Male or Female
- SeniorCitizen – Whether the customer is a senior citizen
- Partner – Whether the customer has a partner
- Dependents – Whether the customer has dependents
- tenure – Number of months the customer has stayed
- PhoneService – Whether the customer has phone service
- InternetService – Type of internet service
- Contract – Type of contract
- PaymentMethod – Payment method used
- MonthlyCharges – Monthly bill amount
- TotalCharges – Total amount charged
- Churn – Whether the customer left the service

3. Tools Used

- **Python** – Data cleaning and Exploratory Data Analysis (Pandas, Matplotlib, Seaborn)
- **SQL** – Business analysis queries
- **Power BI** – Interactive dashboard and data visualization

4. Data Cleaning

During the analysis, some data cleaning steps were required. For example, the Churn column contained hidden characters (carriage return), which caused incorrect churn rate calculation in SQL. These characters were removed using the REPLACE function.

Example SQL cleaning query:

UPDATE churn_data SET Churn = REPLACE(Churn, '\r', '');

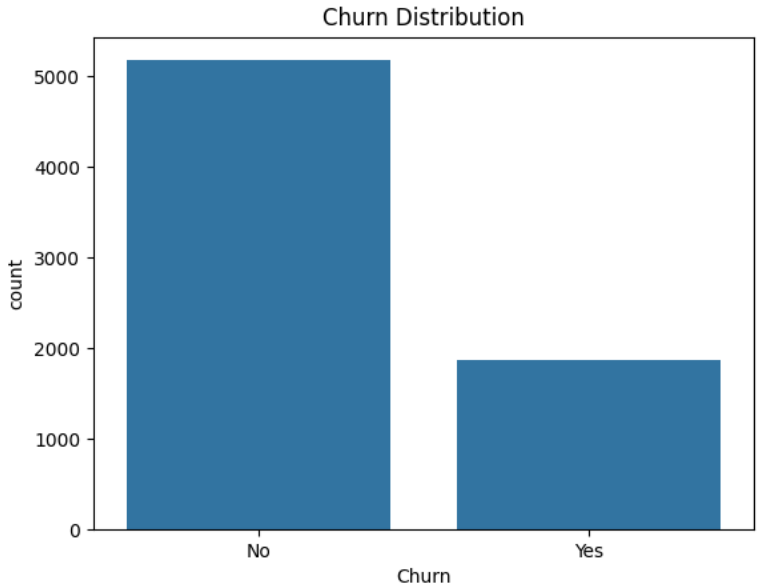
In Python, the TotalCharges column was converted from text to numeric format using pandas to ensure proper calculations.

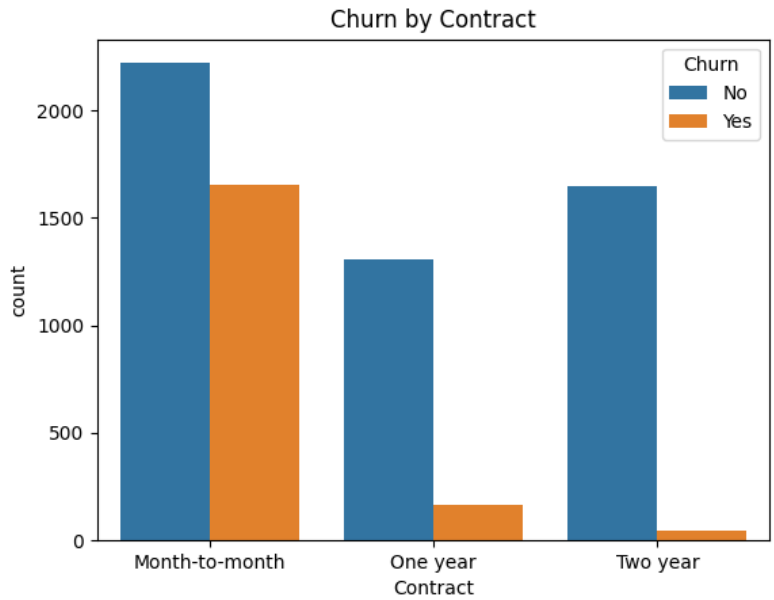
5. Python Exploratory Data Analysis (EDA)

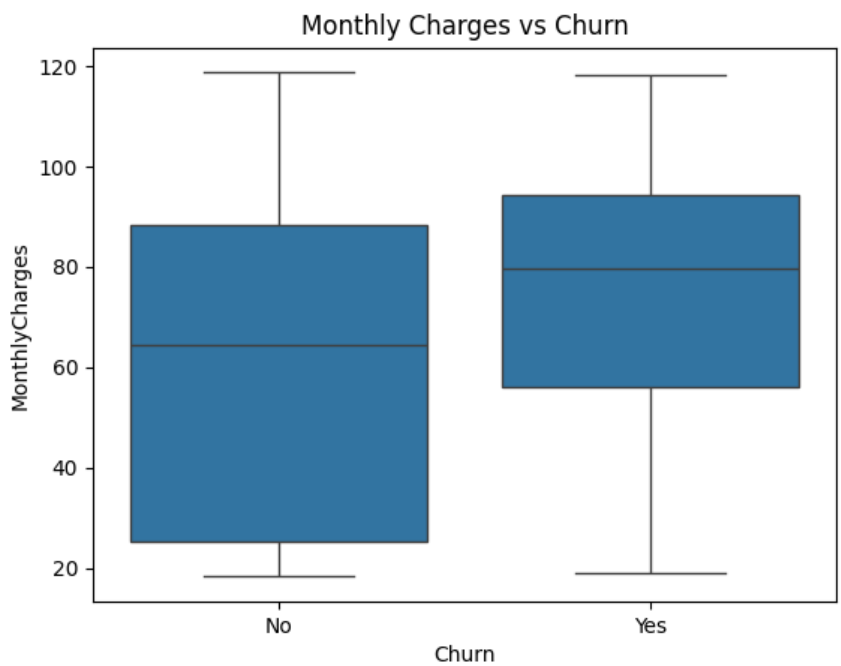
Python was used to explore the dataset visually. The dataset was loaded using pandas, checked for missing values and duplicates, and visualized using seaborn and matplotlib.

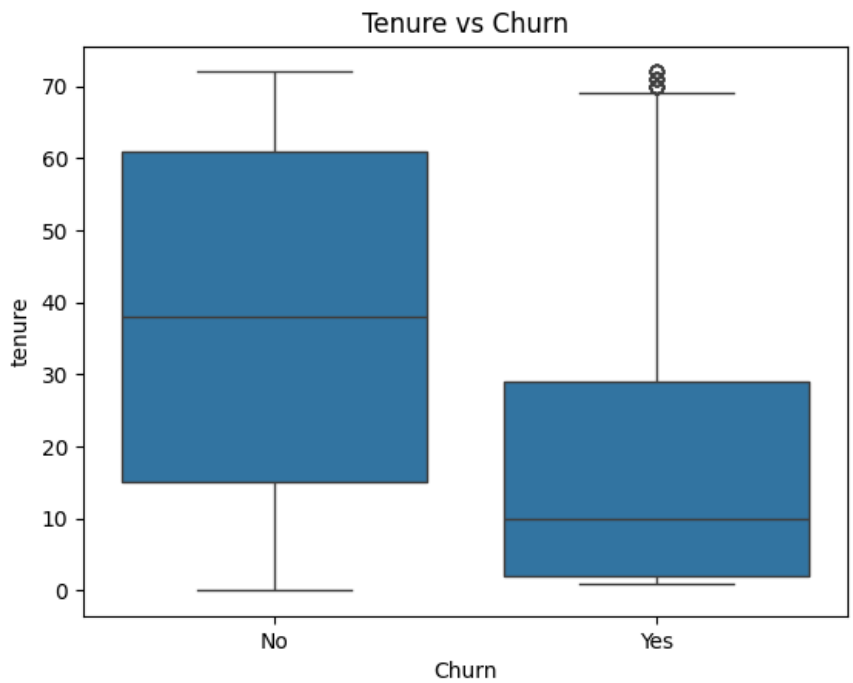
Key visualizations performed:

- Churn Distribution
- Churn by Contract Type
- Monthly Charges vs Churn
- Tenure vs Churn




Churn by Contract
Churn
No
Yes
count
2000
1500
1000
500
0
Month-to-month
One year
Two year
Contract


Monthly Charges vs Churn
MonthlyCharges
120
100
80
60
40
20
No
Yes
Churn



6. SQL Analysis

Several SQL queries were used to analyze the dataset:

1. Churn Rate Calculation

*SELECT ROUND(SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END)*100/COUNT(*),2) AS churn_rate FROM churn_data;*

```
+------------+
| churn_rate |
+------------+
|      26.54 |
+------------+
1 row in set (0.007 sec)
```

2. Customer Distribution by Contract

SELECT Contract, COUNT(*) AS customers FROM churn_data GROUP BY Contract;

Contract	customers
Month-to-month	3875
One year	1473
Two year	1695

3 rows in set (0.021 sec)

3. Average Monthly Charges by Churn

SELECT Churn, AVG(MonthlyCharges) AS avg_monthly_charges FROM churn_data GROUP BY Churn;

Churn	avg_monthly_charges
No	61.26512372083631
Yes	74.44133227059011

2 rows in set (0.011 sec)

4. Churn by Payment Method

SELECT PaymentMethod, COUNT(*) FROM churn_data WHERE Churn='Yes' GROUP BY PaymentMethod;

PaymentMethod	churn_customers
Electronic check	1071
Mailed check	308
Bank transfer (automatic)	258
Credit card (automatic)	232

4 rows in set (0.013 sec)

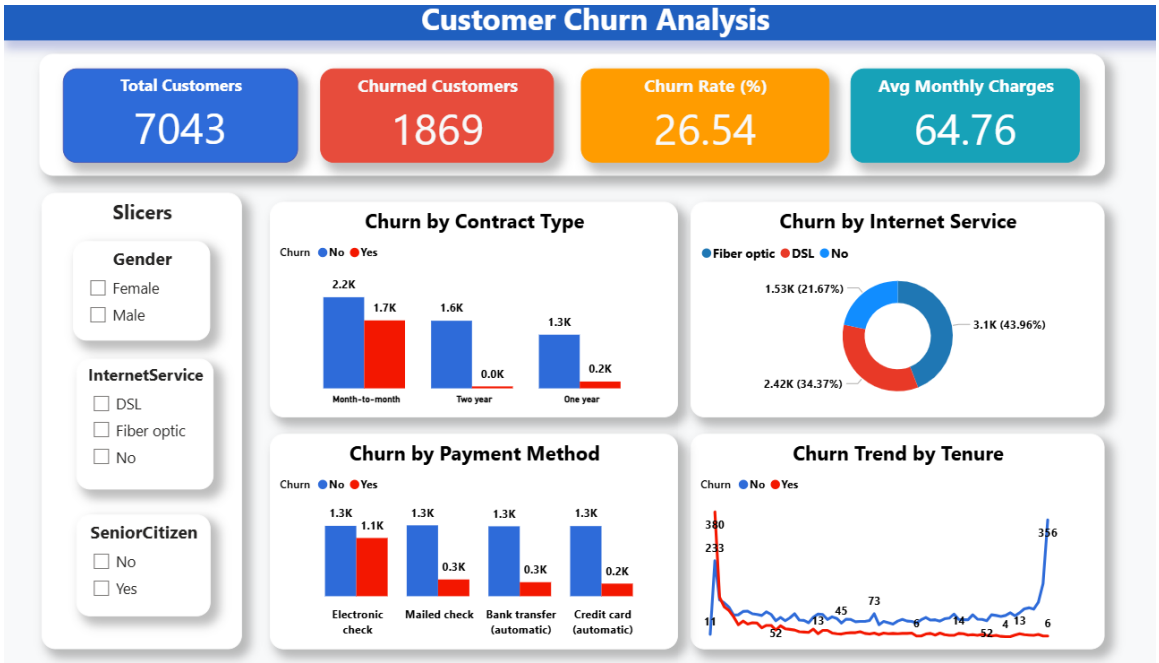
5. Average Tenure by Churn

SELECT Churn, AVG(tenure) AS avg_tenure FROM churn_data GROUP BY Churn;

```
+-------+------------+
| Churn | avg_tenure |
+-------+------------+
| No    |    37.5700 |
| Yes   |    17.9791 |
+-------+------------+
2 rows in set (0.016 sec)
```

7. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



8. Key Insights

- The overall churn rate is approximately 26.5%.
- Customers with month-to-month contracts have the highest churn rate.
- Customers who churn have higher average monthly charges.
- Customers who stay longer (higher tenure) are less likely to churn.
- Electronic check payment method has the highest number of churned customers.

8. Conclusion

The analysis shows that contract type, payment method, monthly charges, and tenure are key factors affecting customer churn. Businesses can reduce churn by encouraging long-term contracts, improving customer support services, and targeting high-risk customers with retention strategies.